



M_ISSION: a MATLAB-based software for multi-GNSS ionospheric scintillation index calculation and analysis

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Abstract

The study of ionospheric scintillation factors for geodetic-type receivers is one of the key approaches to achieving dense ionospheric scintillation monitoring. While several geodetic-type ionospheric scintillation factors have been developed, the lack of open-source platforms for calculating these factors limits their promotion and application. To address this, this paper develops a Matlab-based software for multi-GNSS ionospheric scintillation index calculation and analysis, named as M_ISSION. The software is designed for geodetic GNSS receivers with 30s-sampling interval. The main features of this software are as follows. (1) It supports the calculation of five scintillation factors, including $\sigma_{\phi f}$, ROTI, IAATR, AATR, and DIXSG. (2) It can process the observations from four GNSS systems, i.e. GPS, GLONASS, Galileo, and BeiDou. (3) It can perform temporal and spatial disturbance analysis based on scintillation factor calculation results. The software has a clear interface and is user-friendly. To verify the accuracy and precision of this software, long term GNSS observations are collected from the high-latitude region of Canada, with scintillation monitoring results from ionospheric scintillation monitoring receiver (ISMR) as a reference. The accuracies of the $\sigma_{\phi f}$, ROTI, IAATR, and AATR factors calculated by the software are evaluated and proved with the following aspects, i.e. the scintillation amplitude, the scintillation event occurrence rate. Furthermore, the DIXSG index, derived from the software, is analyzed by comparing it with the Interplanetary Magnetic Field (IMF) Bz component, the Dst index, and the regional ROTI grid. This analysis aims to validate its effectiveness in representing the spatio-temporal characteristics of regional ionospheric disturbances. This software is expected to provide technical support for the promotion and application of various geodetic-type scintillation factors.

Keywords Multi-GNSS · Ionospheric scintillation factors · ROTI · Open-source software

Introduction

Dense and accurate monitoring of ionospheric scintillation is crucial for understanding the disturbance patterns and driving mechanisms of the ionosphere. It is also of significant importance for improving the reliability and robustness

of GNSS positioning and communication systems. Research on ionospheric scintillation factors for geodetic-type receivers is one of the key approaches to achieving dense monitoring of ionospheric scintillation. In recent years, numerous scholars have proposed various ionospheric scintillation factors suitable for geodetic-type receivers. Pi et al. (1997) proposed the Rate of Change of Total Electron Content (TEC) Index (ROTI), which has become one of the most widely used ionospheric scintillation factors (Zhao et al. 2022a). Sanz et al. (2014) introduced the Along Arc TEC Rate factor (AATR), which is believed to be sensitive to regional ionospheric disturbances. Nguyen et al. (2019) and Zhao et al. (2022c) implemented phase scintillation factor extraction from GNSS observations at 1-s and 30-s sampling intervals respectively. Jakowski et al. (2012) and Wilken et al. (2018) proposed the Disturbed Ionosphere Index Spatial Gradient (DIXSG), providing a quantitative standard for global

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ionospheric disturbance evaluation. Luo et al. (2020) proposed the amplitude scintillation index (S_{4c}), significantly expanding the application of geodetic-grade receivers in studying ionospheric irregularities and scintillation phenomena within low-latitude regions.

Currently, there is a lack of an open-source software platform capable of integrating multiple ionospheric scintillation factors from geodetic-type receivers, which indirectly limits the widespread use of these factors and restricts the broader application of such receivers in ionospheric scintillation monitoring. To address this gap, we develop a Matlab-based software for ionospheric scintillation factor calculation and analysis based on geodetic-type receivers, named as M_ISSION. The primary function of this software is to calculate the ROTI, IAATR, AATR, DIXSG, and $\sigma_{\phi f}$ factors based on 30-s observation interval data. It supports batch processing of data from four GNSS systems, i.e. GPS, GLONASS (GLO), Galileo (GAL), and BeiDou (BDS), and can generate regional ionospheric disturbance maps based on scintillation monitoring results. In the following sections, we will briefly introduce the calculation methods for the ionospheric scintillation factors supported by M_ISSION, detail the software implementation process, and evaluate the accuracy of the ionospheric factors generated by M_ISSION.

Overview of ionospheric scintillation factor calculation methods

Data preprocessing

Before calculating the scintillation factors, the raw GNSS data needs to be preprocessed. The satellite cutoff elevation angle is set to 30° to avoid the multipath effect. Observation arcs less than ten epochs are removed. Cycle slip detection and correction are performed using the TurboEdit method (Lichtenegger and Hofmann-Wellenhof 1990), which achieves cycle slip detection by constructing geometry-free (GF) and Hatch-Melbourne-Wübbena (HMW) combinations, as follows,

$$L_{GF} = \lambda_1 \phi_1 - \lambda_2 \phi_2 \quad (1)$$

$$N_{WL} = (\phi_1 - \phi_2) - \frac{f_1 P_1 + f_2 P_2}{f_1 + f_2} \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right) \quad (2)$$

where ϕ_1 and ϕ_2 are the carrier phase observations on the L_1 and L_2 frequencies, respectively, in units of cycles; P_1 and P_2 are the pseudorange observations on the L_1 and L_2 frequencies, in units of meters. f_1 , f_2 and λ_1 , λ_2 are the

frequencies (in units of Hz) and corresponding wavelengths (in units of meters) for the L_1 and L_2 carriers.

Then, epoch differences are computed for the GF and HMW combined observations, denoted as ΔL_{GF} and ΔN_{WL} , respectively. In the absence of cycle slips, these values should exhibit only minor fluctuations around zero. The threshold for the ΔN_{WL} detection value is determined using an adaptive statistical method based on a sliding window (Liu 2011), which calculates the mean and standard deviation of the MW epoch-difference combination values within the window. ΔN_{WL} is identified as a cycle slip event when it exceeds 5 times of the standard deviation. The ΔL_{GF} detection value is susceptible to variations in ionospheric delay. Its threshold is determined using the method proposed by Deo and El-Mowafy (2015). This method employs a function related to the satellite elevation angle to mitigate the significant increase in multipath effects and noise at low elevation angles. A cycle slip is declared for the epoch when the absolute value $|\Delta L_{GF}|$ or $|\Delta N_{WL}|$ exceeds their respective predetermined thresholds.

Once a cycle slip is detected, the following equation is established based on the linear relationship between the magnitude of the combined observations and the integer cycle slips, ΔN_1 and ΔN_2 , on the original carrier phases ϕ_1 and ϕ_2 ,

$$\begin{bmatrix} \Delta L_{GF} \\ \Delta N_{WL} \end{bmatrix} = \begin{bmatrix} \lambda_1 & -\lambda_2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \Delta N_1 \\ \Delta N_2 \end{bmatrix} \quad (3)$$

By solving this system of equations using the least-squares method, float solutions for the cycle slips ΔN_1 and ΔN_2 can be obtained. Rounding these values then yields integer cycle slip estimates. Finally, the computed integer cycle slip values are applied to correct the phase observations starting from the current epoch and all subsequent epochs. This completes the cycle slip repair process, ensuring the continuity of the data arcs.

$\sigma_{\phi f}$ factor

Ghobadi et al. (2020) proposed a method for extracting the phase scintillation index, $\sigma_{\phi f}$, specifically tailored for conventional geodetic receivers with 1 Hz sampling rates and rapidly fluctuating clocks. To enhance the applicability of this index, our software extends this method to datasets featuring a much lower sampling rate of 1/30 Hz. The specific data processing workflow is illustrated in Fig. 1. Compared to 1 Hz sampled data, processing data with a 30-s sampling interval poses greater challenges in handling observation errors and removing trend components. Consequently, optimizations were implemented in these aspects, as detailed below.

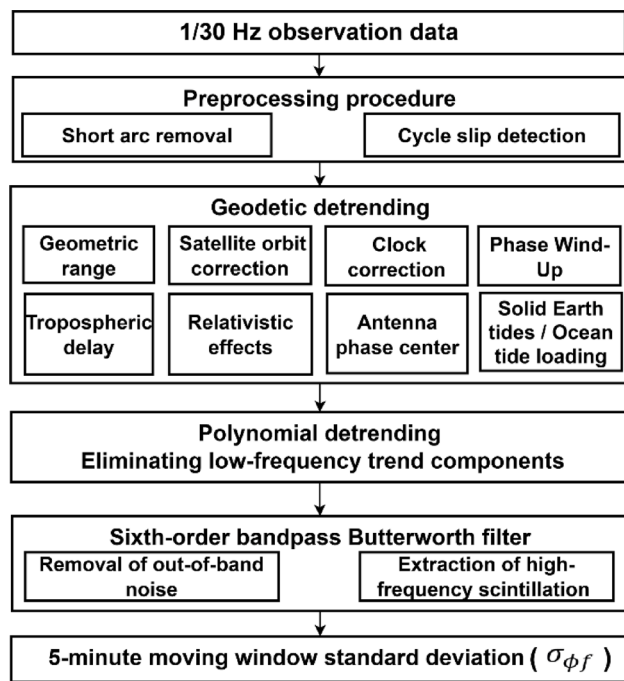


Fig. 1 Flowchart for extracting the phase scintillation index ($\sigma_{\phi f}$) from low-rate geodetic receiver data

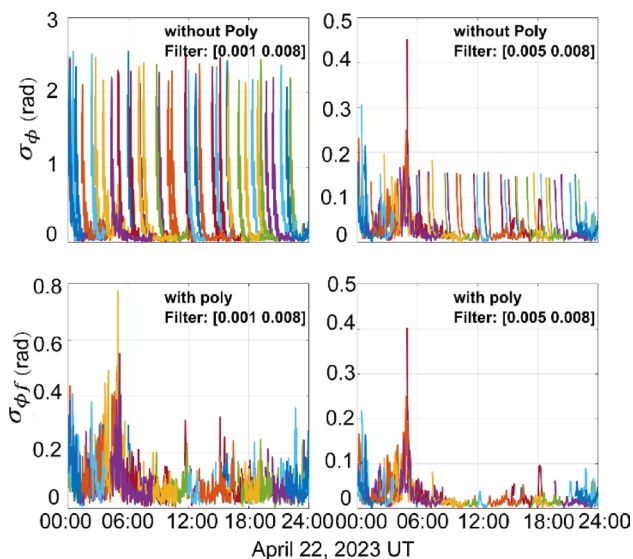


Fig. 2 Effectiveness of the proposed polynomial detrending method in reducing outliers from Butterworth filtering across different frequency bands

First, the geodetic detrending is applied to the GNSS carrier phase observations as follows. Precise satellite ephemeris and clock offset products provided by the International GNSS Service (IGS) are used to correct satellite orbit errors and clock biases. The line-of-sight (LOS) distance between the GNSS satellite and the ground station is derived using precise ephemeris and the station's static position. Station coordinates are sourced from either a user-provided text file

with precise coordinates, Precise Point Positioning (PPP), or the approximate coordinates in the RINEX file header. When coordinates are not provided by the user, our software calculates them using the ionosphere-free PPP technique (Zumberge et al. 1997; Ibanez et al. 2018). The tropospheric delay is corrected using the Saastamoinen model (Saastamoinen 1972). The GPS antenna phase center variations are calibrated through implementation of the IGS absolute antenna model. The relativistic effect is corrected by employing Kouba's (2015) theoretical formulation. The phase wind-up compensation is estimated by the algorithm of Wu et al. (1993), while the solid Earth and ocean loading effects are modeled by the framework proposed by Petit and Luzum (2010). For the data with 1 s sampling interval, Ghobadi et al. (2020) adopted a sixth-order high-pass Butterworth filter to remove ionospheric refraction effects and low-frequency noise in the phase signal residual after geodetic detrending. In contrast, a more refined sixth-order Butterworth band-pass filter was employed here to remove residual low-frequency noise and high-frequency interference, thereby isolating the high-frequency ionospheric scintillation component. However, for data sampled at 30-s interval, low-frequency residual trends are more prone to remain after the initial geodetic detrending. The conventional Butterworth band-pass filter is highly sensitive to such low-frequency trends in the data. These trends will interfere with determining the scintillation frequency band interval in low-sampling-rate data (Zhao et al. 2023), causing anomalous scintillation index values. The upper two panels of Fig. 2 illustrate phase scintillation indices extracted directly from detrended 30-s data at station CHUC on April 22, 2023, using Butterworth filters with [0.001 0.008] Hz and [0.005 0.008] Hz bandwidths. The results exhibit significant anomalous fluctuations characterized by apparent trend-like patterns.

To address this issue, an additional step is introduced prior to the Butterworth filtering by applying an arc-by-arc third-order polynomial method to eliminate the residual low-frequency trend term, as follows,

$$P(t) = a_3 t^3 + a_2 t^2 + a_1 t + a_0 \quad (4)$$

where t denotes the sample time, and (a_3, a_2, a_1, a_0) represent the polynomial coefficients. Employing the least-squares criterion, the polynomial coefficients are determined by minimizing the sum of squared residuals between the original signal and the polynomial model. Finally, this fitted optimal trend polynomial is subtracted from the original signal to yield a detrended signal sequence, providing a suitable foundation for subsequent filtering analysis. The lower two panels of Fig. 2 display the extracted scintillation indices after removing the low-frequency trend component

using this cubic polynomial approach. These results demonstrate the effective attenuation of the anomalous fluctuations previously observed. This result confirms the capability of the proposed cubic polynomial method to mitigate outliers originating from non-scintillation sources. It provides a clean background signal for Butterworth filtering, reduces the Butterworth filter's sensitivity to the selection of the cutoff frequency, and consequently enhances the accuracy and robustness of the overall scintillation index extraction algorithm.

Finally, a 5-min sliding window is employed to calculate the standard deviation of the carrier phase residuals, yielding the 30 s interval phase scintillation factor (Ghobadi et al. 2020; Zhao et al. 2023), as follows,

$$\sigma_{\phi_f} = \sqrt{\langle \phi_f^2 \rangle - \langle \phi_f \rangle^2} \quad (5)$$

where σ_{ϕ_f} is the phase scintillation factor (Zhao et al. 2022c); ϕ_f represents the carrier phase residuals after cycle slip detection, outlier removal, geodetic detrending and ionospheric refraction removal using polynomial method and a sixth-order Butterworth filter, with the units of in radians; $\langle \cdot \rangle$ denotes the arithmetic mean.

ROTI factor

The ROTI factor (Pi et al. 1997; Zhao et al. 2024) is calculated using the sliding window method, consistent with the σ_{ϕ_f} calculation. The specific calculation steps are as follows,

$$\phi(m) = \phi_1(m) \times \lambda_1 - \phi_2(m) \times \lambda_2 \quad (6)$$

$$\text{ROT}(m) = \frac{\phi(m) - \phi(m-1)}{\Delta t \times 10^{16} \times 40.3 \times \left(\frac{1}{f_1^2} - \frac{1}{f_2^2} \right)} \quad (7)$$

$$\text{ROTI} = \sqrt{\langle \text{ROT}^2 \rangle - \langle \text{ROT} \rangle^2} \quad (8)$$

where m is the epoch index; $\phi(m)$ is the uncombined phase observation value in units of meters; Δt is the time difference between two adjacent epochs in units of minutes; ROT is the rate of change of total electron content.

AATR factor

The specific calculation method of the AATR factor (Juan et al. 2018) is as follows,

$$IAATR_i^j(t) = \frac{1}{(\sec(e))^2} \text{ROT} \quad (9)$$

$$AATR(T) = \sqrt{\frac{1}{N} \sum_{t=T}^{T+\Delta T} \sum_{j=1}^{n_{sat}(t)} \left(IAATR_i^j(t) \right)^2} \quad (10)$$

where $IAATR_i^j(t)$ is the instantaneous AATR parameter between satellite j and receiver i at time t ; e is the satellite elevation angle; $\sec(e)$ is the slant path factor; ΔT is set to a duration of one hour; n_{sat} is the number of visible satellites during that time period; $AATR(T)$ is the arc TEC rate at time T .

DIXSG factor

The method for calculating DIXSG is as follows (Wilken et al. 2018). First, the gradient variation of ROT between adjacent piercing points and its difference between receivers are calculated,

$$c\text{ROT}_i^k = \frac{\text{ROT}}{\Delta S_i} \quad (11)$$

$$\text{DIXSG}(c\text{ROT}_{(\text{level})})_{i,j}^k = \left(\frac{|c\text{ROT}_i^k - c\text{ROT}_j^k|}{c\text{ROT}_{(\text{level})}} \right)^3 \left(\frac{d}{D} \right)^{-1} \quad (12)$$

where k is the satellite; Δs is the distance between the ionospheric piercing points at two different times; i and j are the ground receivers; $c\text{ROT}_{(\text{level})}$ represents the selectable identification level; d is the distance between satellite k and the corresponding ionospheric piercing points of receivers i and j , limiting between 10 and 1000 km; D is the maximum allowable distance, set as $D = 1000$ km. Then, a simplification is applied, where values of $\text{DIXSG}(c\text{ROT}_{(\text{level})}(E))$ greater than 1 are defined as 1, otherwise as 0. Finally, the arithmetic mean is calculated to obtain the global DIXSGp,

$$\text{DIXSG}_{(n-\text{Level})} = \sum_{E=1}^n \text{DIXSG}(c\text{ROT}_{(\text{level})}(E)) \quad (13)$$

$$\text{DIXSGp} = \frac{1}{N} \cdot \sum \text{DIXSG}_{(n-\text{Level}, \max(l^\circ \times b^\circ))} \quad (14)$$

where n is the number of sensitivity levels; E is the corresponding level; N is the number of valid grid points; l° and b° are the latitude and longitude of the divided grid unit, respectively.

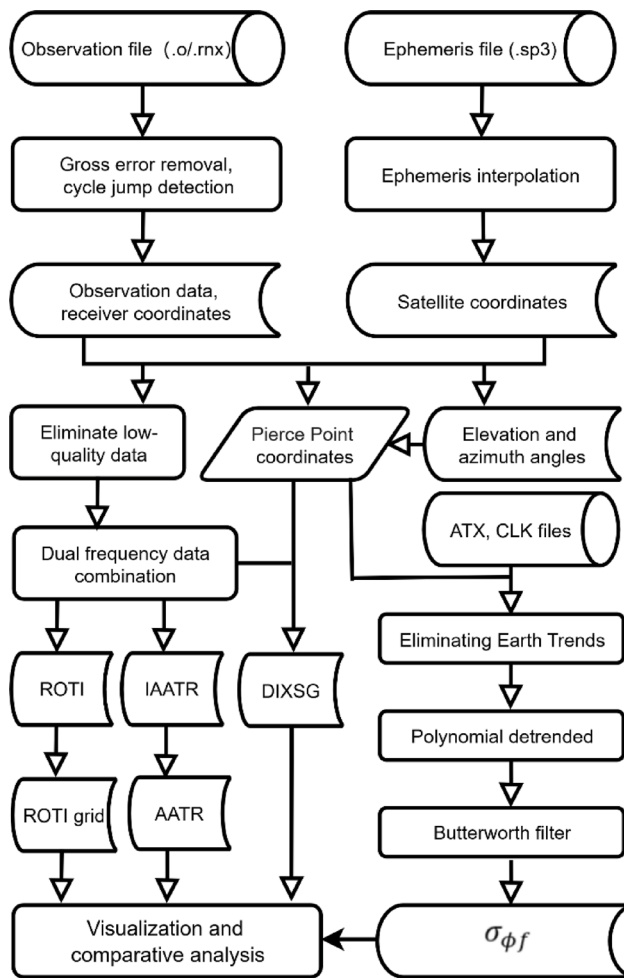


Fig. 3 Implementation flow of M_SSION software

Software design

M_SSION was developed utilizing a MATLAB-based graphical user interface (GUI) programming framework. The software was developed and tested on MATLAB R2023b and R2024a platforms; users are recommended to employ MATLAB R2023b or later releases to ensure operational stability. M_SSION comprises three functional interfaces, i.e. ionospheric scintillation index calculation, scintillation index visualization, and scintillation index analysis. The ionospheric factor calculation interface allows the calculation of factors such as ROTI, AATR, DIXSG, and $\sigma_{\phi f}$. It also includes common parameter settings, such as configuring the paths for observation data and precise ephemeris data, system selection, setting the cutoff elevation angle, and configuring cycle slip detection parameters. Additionally, it allows for customization of DIXSG sensitivity, grid size, latitude and longitude ranges, as well as the clock bias and antenna phase center file paths for $\sigma_{\phi f}$, and the filtering threshold.

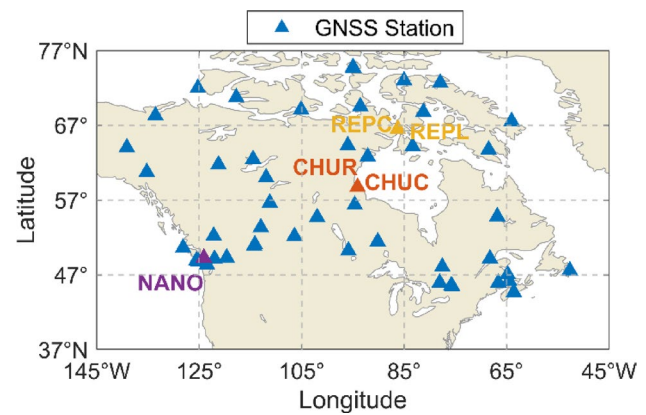


Fig. 4 Distribution of the stations used in this paper

The scintillation factor visualization interface mainly focuses on the analysis of scintillation factor calculation results for specified satellites. The scintillation factor analysis interface primarily compares and analyzes the time series accuracy of ROTI, AATR, DIXSG, and $\sigma_{\phi f}$ factors, and supports the 2D gridding of ROTI and DIXSG factors. The software implementation flow is shown in Fig. 3. Some of the functionalities were referenced from the M_GIM (Zhou et al. 2023), PPPH (Bahadur and Nohutcu 2018), and GLAB (Ibanez et al. 2018).

M_SSION is suitable for data from geodetic-type receivers with a 30s sampling interval, compatible with both Rinex 3 and Rinex 2 versions. It should be noted that data with a 30-s sampling interval, compared to those with 1-s or higher-resolution intervals, may be insufficient to capture extremely short-duration ionospheric scintillation. This limitation poses greater challenges for obtaining accurate scintillation results and demands more stringent data processing requirements. However, since nearly all IGS stations provide 30-s interval data, considering their broad distribution and easy accessibility, which facilitates global ionospheric scintillation monitoring, our software therefore prioritizes support for 30-s data over 1-s or higher-resolution formats. M_SSION supports fast output and verification of intermediate files such as cROT, elevation angle, and azimuth angle. It also supports batch calculations of ROTI, AATR, DIXSG ionospheric scintillation factors for GPS, Galileo, GLONASS, and BeiDou systems.

Experiment and results discussion

Data description

To verify the monitoring performance of the M_SSION software, observations from the Canadian region were collected from the stations, as shown in Fig. 4. This dataset

includes 24 stations from the Canadian High Arctic Ionospheric Network (CHAIN) and 33 stations from IGS, with a sampling interval of 30 s. Among these, the CHUR station is equipped with a Septentrio PolaRx5 receiver, while its adjacent CHUC station is equipped with a Septentrio PolaRx5S ionospheric scintillation monitoring receiver (ISMR). The distance between the CHUC and CHUR is approximately 124 m. Similarly, the REPC station is equipped with a Septentrio PolaRx5S receiver, while the REPL station features a TRIMBLE NETR9 receiver. REPC and REPL are co-located stations. The NANO station supports observations from four GNSS systems.

This study collected data during the geomagnetic storm event occurred from April 22 to 24, 2023. This event was the largest geomagnetic disturbance in the 25th solar cycle up until that time. The geomagnetic condition is shown in Fig. 5. The ring current index, i.e. Dst, gradually decreased starting from 12:00 Coordinated Universal Time (UTC) on the 23rd, while the global geomagnetic activity index, namely Kp, gradually increased during the same period. Two peaks were observed at 21:00 UTC on the 23rd and 06:00 UTC on the 24th. After these peaks, the values gradually returned to a calm level. The geomagnetic storm lasts for over 24 h.

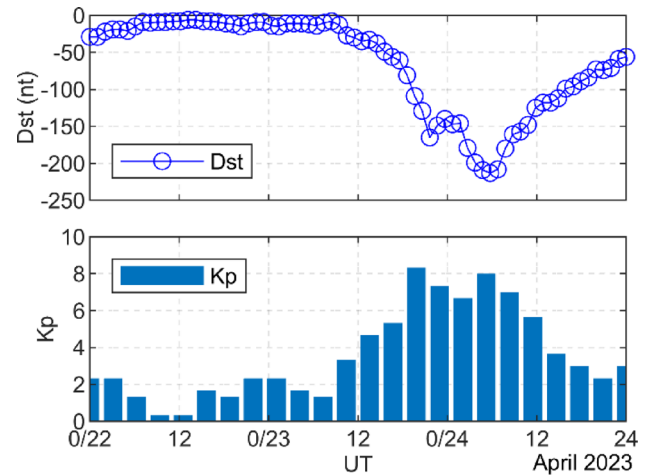
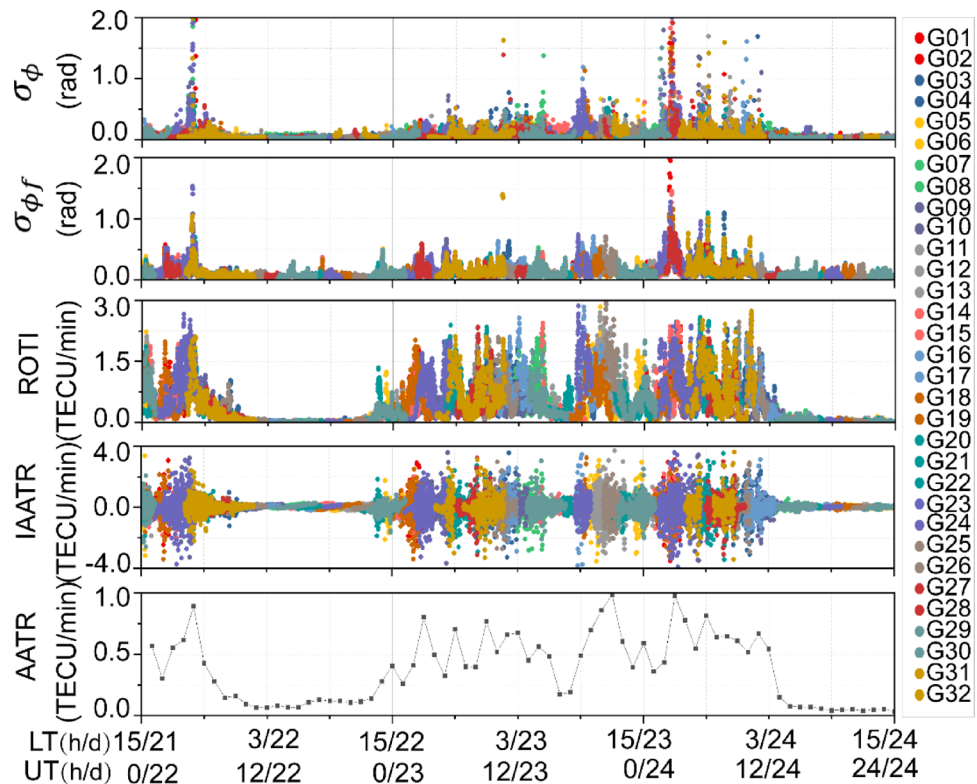


Fig. 5 Global Dst and Kp from April 22 to 24, 2023

Accuracy analysis of ionospheric scintillation monitoring with $\sigma_{\phi f}$, ROTI, IAATR, and AATR

This paper utilizes phase scintillation data (σ_{ϕ}) obtained directly from the ISMR receiver at station CHUC as reference values to assess the accuracy of $\sigma_{\phi f}$, ROTI, IAATR and AATR indices derived from the geodetic receiver at the adjacent CHUR station. The results are shown in Fig. 6. By comparing with σ_{ϕ} , it can be observed that the magnitudes of ROTI, IAATR, and AATR exhibit scintillations during the same disturbed periods given by σ_{ϕ} , with the overall

Fig. 6 Comparison of σ_{ϕ} from the CHUC station and various geodetic scintillation factors calculated at the adjacent CHUR station, with Universal Time (UT) and Local Time (LT) scales annotated in the hour(h)/day(d) format



variation trend similar to σ_ϕ as well. However, these three factors may still issue strong scintillation warnings, even during the calm period of σ_ϕ . The primary reason for the excessive false positives is that all three scintillation indices are calculated based on TEC, causing them to struggle in differentiating between TEC gradient-induced perturbations and authentic scintillation signatures (Wang et al. 2021). The auroral oval region where the CHUR station is located exhibits intense electron density gradient variations, which makes the three TEC-based scintillation indices prone to false positives in scintillation detection. The fluctuations in IAATR reflect the rapid variations in the ionospheric density gradient along the arc direction, with its fluctuations synchronized with the changes in ROTI. AATR, due to its one-hour time resolution, exhibits a longer duration of scintillation characteristics. The errors between the ROTI, IAATR, and AATR factors and the reference value align with previous studies (Zhao et al. 2022a). From Fig. 6, it can be observed that the results of σ_ϕ and $\sigma_{\phi f}$ exhibit good consistency. Both factors detect strong scintillation around 05:00 UTC on the 22nd and 03:00 UTC on the 24th. Multiple moderate scintillation events are observed throughout the 23rd and before 12:00 UTC on the 24th. Results reveal an overall tendency for $\sigma_{\phi f}$ values to be greater than σ_ϕ measurements. This may be attributed to the higher background noise present in the 30-s sampling interval data, which may not be eliminated by the detrending and filtering operations. Consequently, $\sigma_{\phi f}$ values typically surpassed σ_ϕ during sporadic strong scintillation, most moderate-to-weak events, and quieter intervals. However, during strong scintillation events, σ_ϕ frequently exceeded $\sigma_{\phi f}$. This reversal is likely an effect of the chosen Butterworth band-pass filter cutoff frequencies, which are set as [0.001, 0.015] Hz.

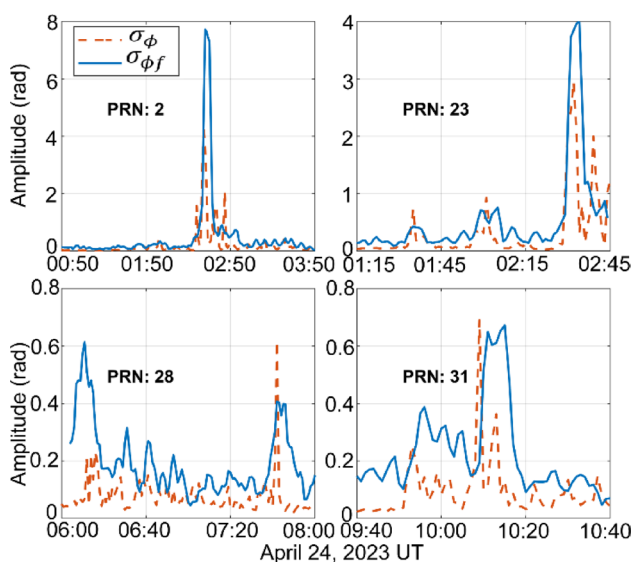


Fig. 7 Comparative analysis of σ_ϕ at CHUC station and $\sigma_{\phi f}$ at CHUR station on 24 April 2023 in different arc segments

It is also possible that the low data-sampling rate plays a role, as its inherent Nyquist limit could restrict the detection of higher-frequency components. This combination of potential limitations might lead to an underestimation of high-frequency energy during intense scintillation events by $\sigma_{\phi f}$ compared to the reference.

To further examine the characteristics of the $\sigma_{\phi f}$ index, Figure Fig. 7 presents a direct comparison between its derived arcs and the reference σ_ϕ arcs from the ISMR receiver. Observational results indicate that $\sigma_{\phi f}$ typically exhibits higher amplitude values with longer-duration scintillation events than those provided by σ_ϕ , while $\sigma_{\phi f}$ exhibits trend consistency with σ_ϕ . These characteristics primarily arise from intrinsic methodological differences between the two indices. $\sigma_{\phi f}$ is computed using low-rate (1/30 Hz) data with a 5-min moving window, whereas σ_ϕ relies on 50 Hz high-rate data and a shorter 1-min window. Additionally, the clock stability of geodetic receivers is generally lower than that of ISMR, making it more challenging to mitigate receiver clock bias effects during data processing, thereby compromising the accuracy of the scintillation index (Zhao et al. 2023).

While differences exist between $\sigma_{\phi f}$ and σ_ϕ regarding scintillation amplitude and duration, the validity of $\sigma_{\phi f}$ warrants further evaluation through scintillation event detection accuracy (Jiao and Morton 2015). Scintillation events are defined as instances where the scintillation index exceeds a specified threshold for durations exceeding two minutes (Jiao and Morton 2015). The event occurrence rate represents the percentage of hourly scintillation events relative to the total scintillation events recorded during the statistical period, such as April 22 to 24, 2023 UTC, as used in this paper. The threshold for the σ_ϕ index was set at 0.2 rad (Ahmed et al. 2015). For station CHUR, the $\sigma_{\phi f}$ threshold was determined as approximately 0.5 rad using the Complementary Cumulative Distribution Function (CCDF) method (Zhao et al. 2022b). Figure 8 displays scintillation event occurrence rates derived from σ_ϕ measurements of the ISMR receiver at station CHUC and $\sigma_{\phi f}$ calculations based on 30-s sampled data at station CHUR. Good agreement is observed between the occurrence rates identified by σ_ϕ and $\sigma_{\phi f}$. This indicates a relatively high reliability in scintillation event monitoring using the $\sigma_{\phi f}$ index, despite the known amplitude discrepancies between the two indices.

To evaluate the applicability of the $\sigma_{\phi f}$ index across different GNSS receiver types, $\sigma_{\phi f}$ values calculated from station REPL equipped with a TRIMBLE NETR9 receiver were compared with σ_ϕ measurements from station REPC using a Septentrio PolaRx5S receiver. The comparison results are presented in Fig. 9. During periods of strong ionospheric scintillation, exemplified as Shadow 1, the dynamic variations of $\sigma_{\phi f}$ exhibit good consistency with the reference

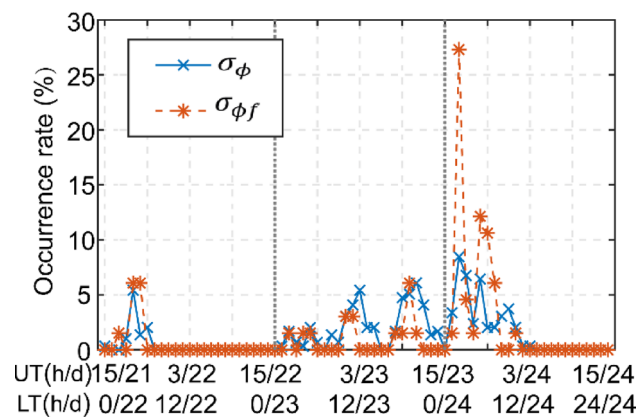


Fig. 8 Comparison of scintillation occurrence rates between σ_ϕ and $\sigma_{\phi f}$

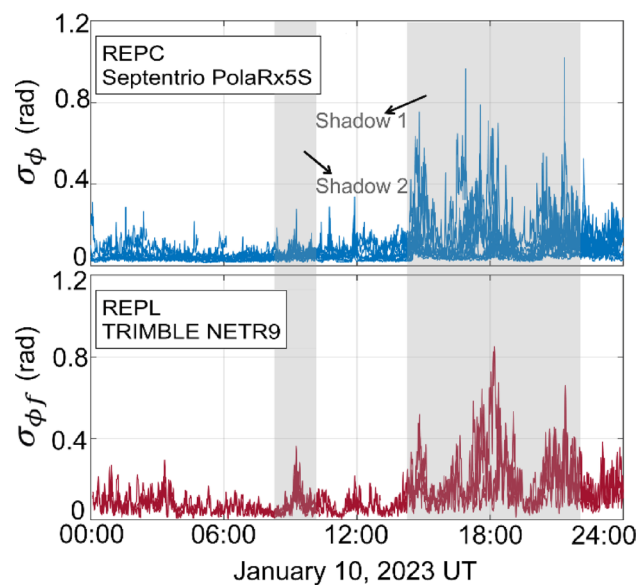


Fig. 9 Comparison of σ_ϕ from the REPC station and $\sigma_{\phi f}$ at the co-located REPL station

index σ_ϕ , although there are differences in magnitude. lower. In contrast, during some intervals of weak ionospheric scintillation, such as the one highlighted in Shadow 2, the $\sigma_{\phi f}$ values are significantly higher than the reference values. In addition to the inconsistency between σ_ϕ and $\sigma_{\phi f}$ discussed in the context of Fig. 7, the differentiated performance of various conventional receiver types when compared with the ISMR reference values may also be attributed to differences in their internal oscillator performance. It should also be noted that the calculation of $\sigma_{\phi f}$ is dependent on a band-pass filter whose cutoff frequency requires adjustment based on station-specific conditions, and that minor variations in this frequency can influence the final magnitude of the index. Nevertheless, the existence of these discrepancies does not undermine the reliability of the $\sigma_{\phi f}$ index as an effective indicator for detecting strong scintillation events.

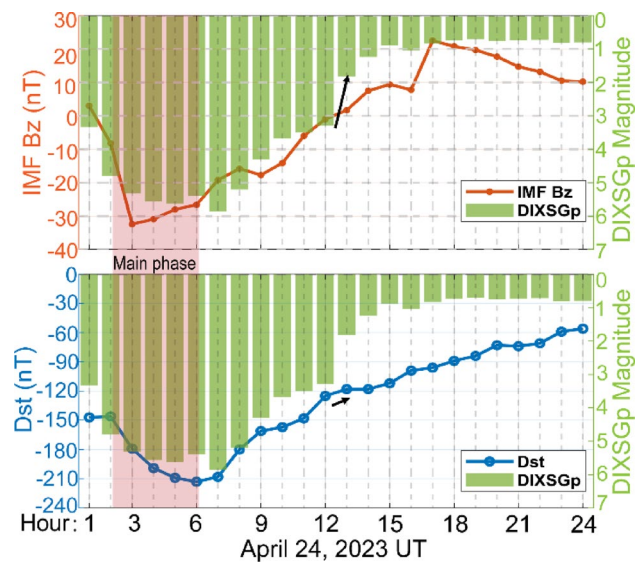


Fig. 10 Comparison of DIXSGp over the Canadian region with the Interplanetary Magnetic Field (IMF) Bz component and Dst index on 24 April 2023. The top panel depicts temporal variations of DIXSGp versus IMF Bz, while the bottom panel displays DIXSGp variations relative to the Dst index

While some studies also highlight the robust stability of Trimble receivers during ionospheric scintillation (Jia et al. 2024), we incorporated this receiver primarily to demonstrate the broader applicability of our method. Due to experimental constraints, a systematic validation on other types of receivers was not performed.

Accuracy analysis of the DIXSG index

To comprehensively evaluate the capability of DIXSGp in characterizing the global ionospheric state during a severe geomagnetic storm, the event of April 24, 2023, was selected for analysis. A synchronous comparison was performed among the DIXSGp index, the interplanetary magnetic field (IMF) Bz component, and the geomagnetic Dst index. As depicted in Fig. 10, the IMF Bz component witnessed a sharp southward turning at approximately 02:00 UT, creating conditions for sustained energy injection into the Earth system (Burton et al. 1975). In response, the DIXSGp index, previously at quiet levels, increased rapidly to a peak of 5.86 at 07:00 UT. Concurrently, the Dst index dropped sharply to a minimum of -213 nT at 06:00 UT. The variations among these three parameters exhibited a high degree of consistency during the main phase of the storm, thereby validating the effectiveness of DIXSGp in capturing the intensity of global ionospheric disturbances. However, the key difference in the response characteristics between DIXSGp and Dst became apparent during the recovery phase following the main phase, as indicated by the arrow in the figure. At approximately 12:00 UT, when

the IMF Bz turned northward and the energy injection diminished (Akasofu 1981), the DIXSGp index decreased rapidly, demonstrating a real-time response to the change in the external driver. In contrast, the Dst index representing ring current energy exhibited a more gradual recovery, consistent with energy dissipation governed by slower physical processes (Gonzalez et al. 1994). This divergence in temporal response characteristics demonstrates DIXSGp's capacity to characterize both immediate energy input effects and, to some extent, slowly evolving energy storage effects in the magnetosphere. This analytical approach, similar to the method by Wilken et al. (2018), confirms that DIXSGp accurately and sensitively captures global or regional ionospheric disturbances triggered by abrupt interplanetary condition changes.

Additionally, to validate the effectiveness of DIXSG for characterizing localized regional ionospheric disturbances, the well-established ROTI data was used as a reference. A spatial grid for ROTI was constructed wherein each grid value is defined as the spatially averaged ROTI. This approach enables a quantitative assessment of the performance and reliability of DIXSG for monitoring regional

ionospheric disturbances by confirming the spatiotemporal consistency between the two datasets. The DIXSG sensitivity parameter is configured with eight discrete levels, incrementally increasing from 50 to 400. As shown in Figs. 11 and 12, both DIXSG and ROTI variations in the Canadian sector on 24 April 2023 exhibit consistent spatiotemporal patterns with the IMF Bz and Dst index fluctuations presented in Figure Fig. 10. During 00:00–06:00 UTC, ionospheric disturbances intensified progressively, expanding from mid-high to mid-low latitudes. Following this period, scintillation intensity gradually diminished, retreating to mid-high latitudes after 06:00 UTC. Post-13:00 UTC, with the IMF Bz turning northward, reduced geomagnetic activity correlated with further suppression of scintillation occurrences, demonstrating spatial coherence. By 18:00 UTC, elevated values predominantly localized near polar regions, attributable to open magnetic field line configurations enhancing ionospheric activity (Schunk and Sojka 1997). The congruent spatial evolution between DIXSG and ROTI, synchronized with geomagnetic index variations, confirms that the M_ISSION-derived DIXSG factor effectively characterizes ionospheric disturbance dynamics.

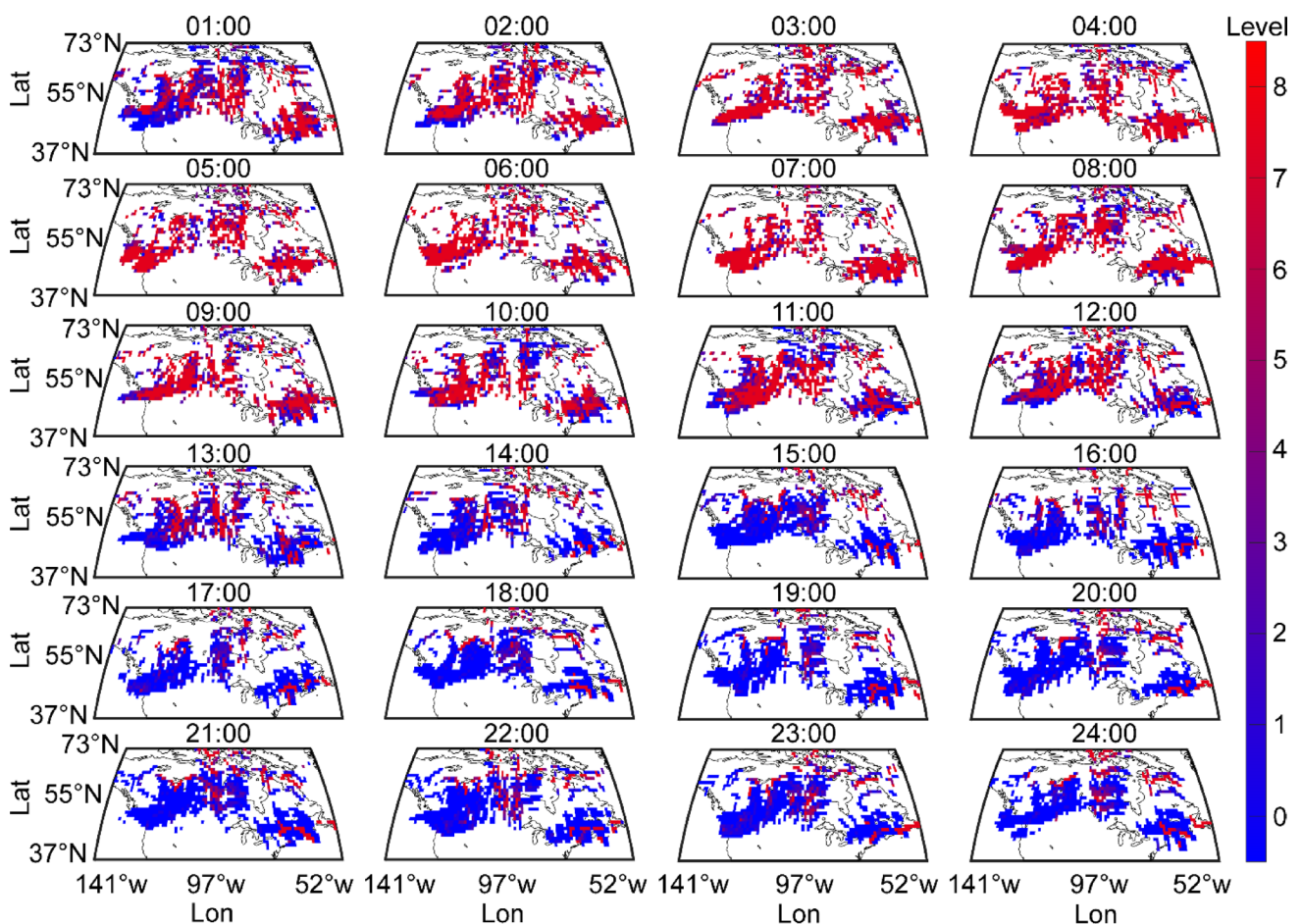


Fig. 11 DIXSGp variation over the Canadian region with longitude (Lon) and latitude (Lat) coverage, from 1:00 to 24:00 UTC on April 24, 2023

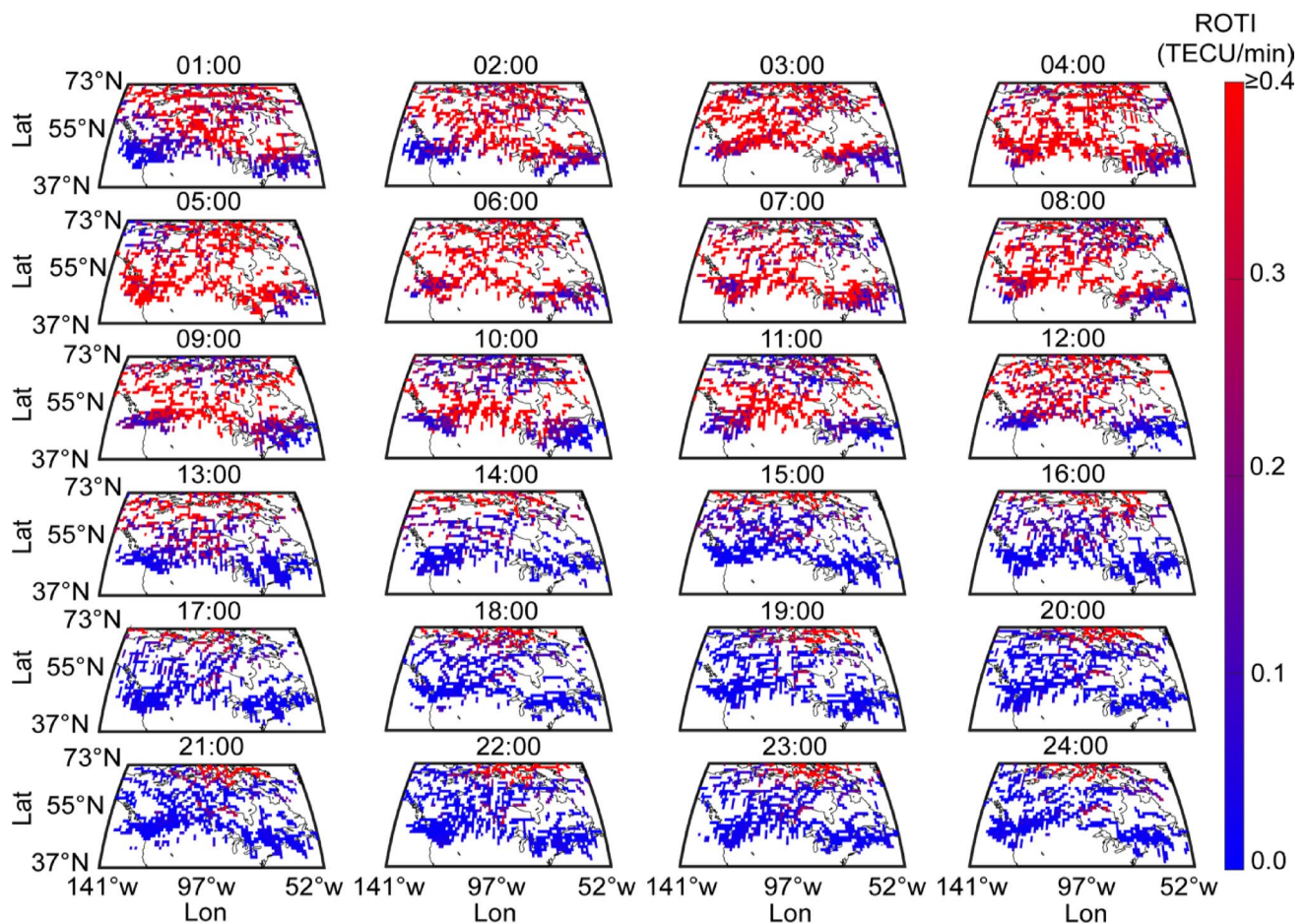


Fig. 12 ROTI variation in the Canadian region from 1:00 to 24:00 (UTC) on April 24, 2023

Conclusion

This study introduces M_SSION, a MATLAB-based software platform designed for computing and analyzing ionospheric scintillation indices. The developed tool processes 30-s interval data to calculate multiple indices including ROTI, IAATR, AATR, and DIXSG for GPS, GLONASS, Galileo, and BeiDou systems, along with $\sigma_{\phi f}$ index specifically for GPS. The software incorporates data visualization capabilities through its graphical user interface, enabling users to load stored results, generate plots, and perform comprehensive analysis. A dedicated computation window guides users through the complete processing workflow while allowing flexible parameter configuration via interactive GUI controls. Due to practical limitations, the current software implementation does not exhaustively include all ionospheric scintillation factors applicable to geodetic receivers. Future development will prioritize the code integration and validation of supplementary factors, with planned updates to be incorporated in subsequent software releases. The M_SSION package, comprising source code, documentation, and sample datasets, is publicly accessible

via the GitHub repository at: https://github.com/DS-Zhao/M_SSION/tree/master.

The ionospheric disturbance monitoring capability of the M_SSION software was evaluated using GNSS data from a geomagnetic storm-affected region in Canada. By implementing an improved outlier removal and a polynomial detrending algorithm, the phase scintillation factor $\sigma_{\phi f}$ was extracted from data with a 30s sampling interval. A comparative analysis with σ_{ϕ} from ISMR subsequently validated the accuracy of the ROTI, IAATR, AATR, and $\sigma_{\phi f}$ products generated by M_SSION. Furthermore, the capability of DIXSG to accurately reflect the spatiotemporal evolution of ionospheric disturbances during geomagnetic storms was verified using ROTI grid data from M_SSION in conjunction with the IMF Bz and Dst index. As the global demand for space weather monitoring increases, M_SSION serves as a specialized tool for ionospheric scintillation analysis, offering support for scintillation index computation and data analysis in fields such as GNSS positioning and space physics research.

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Author contribution D.Z. and P.Z. conceptualized the study, developed the software code, and wrote the original manuscript. X.Z., S.C. and H.L. contributed to software modification and validation. K.Z., Q.W. and C.L. provided critical review, funding acquisition, and supervised the research. All authors reviewed and approved the final manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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